

# SHUBHAJIT ROY

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## Education

### Indian Institute of Technology Gandhinagar

Ph.D in Computer Science and Engineering

CPI - 9.14

July 2022 – Present

Gandhinagar, India

### Indian Association for the Cultivation of Science

M.Sc in Mathematics and Computing with Major in Computer Science

CGPA - 8.42

October 2020 – July 2022

Kolkata, India

### Presidency University

B.Sc in Mathematics

CGPA - 6.95

June 2017 – August 2020

Kolkata, India

## Experience

### Research Intern - Platforms in AI

IBM Research

May 2026 – August 2026

Bangalore, India

## Publications

- “Si’multaneous ’S’patial-’T’emporal Message Passing for Dynamic Graph Representation Learning” ArXiv (2026). Shubhajit Roy and Anirban Dasgupta.
- “FIT-GNN: Faster Inference Time for GNNs that ‘FIT’ in Memory Using Coarsening” TMLR (2026). Shubhajit Roy, Hrriday Ruparel, Kishan Ved and Anirban Dasgupta.
- “Local Fragments, Global Gains: Subgraph Counting using Graph Neural Networks” DiffCoAlg @ NeurIPS (2025), WEB&GRAPH @ WSDM 2026. Shubhajit Roy, Shrutimoy Das, Binita Maity, Anant Kumar, and Anirban Dasgupta.

## Projects

### Graph Representation Learning for Fast Detector Simulation

Prof. Shanmuganathan Raman

January 2024 - April 2024

- Quark and gluon jets from high-energy particle collisions produce distinct patterns in jet images, but distinguishing them remains a core challenge in particle physics.
- This project applies machine learning, especially GNNs, VAEs, and GVAEs, to detect patterns that differentiate quark- from gluon-origin jets.

### Graph Neural Network for weather data prediction

Prof. Anirban Dasgupta

February 2023 - Present

- Given a set of locations as nodes and edges between, our task is to predict different weather stats based on past information.
- We propose to use GNNs in this streaming fashion and still overcome the current problems faced while learning from streaming graphs.
- One major problem tackled is the catastrophic forgetting problem, where the GNN model tends to forget past patterns in the graphs. Another problem faced is the memory consumption of these models.

### Joint Entity Recognition and Relation extraction

Dr. Partha Basuchowdhuri

September 2021 - Present

- Given a sentence, we need to find the entities in the sentences and the relation between these entities. This task helps us in the construction of a Knowledge Graph where the nodes are the entities, and the edges between these nodes have the corresponding relation.
- In our proposed model, we intend to use GNNs to extract the dependencies between the words in a sentence and then predict the entities and relations.

### MedEx - A Software for an Pharmacy database

Prof. Paramartha Dutta

June 2021 - July 2021

- Implemented the SRS and the prototype code for transaction management of an online pharmaceutical database.
- The software helps the customer of the wholesaler to open an account and do all the transactions in a digital method. It includes facilities like credit payment, seeing transaction history, etc.
- Done along with Sarthak Das and Avishek Lahiri as a part of the IACS Computer Science Unit 2020 batch.

March 2021 - August 2021

- A tool for verification of ReLU neural network (for example, Acas-Xu) properties.
- The tool uses SCIP for optimizing and finding the bound with respect to the neural network and z3 for checking satisfiable conditions.
- Done along with Sarthak Das and Avishek Lahiri as a part of the IACS Computer Science Unit 2020 batch.

## Roles and Responsibilities

|                                                                                                                                                                            |                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
|  <b>Tutorial - ACM Summer School: AI for Social Good (Graph Neural Networks: Hands On)</b> | <b>June 2025</b>                   |
| Indian Institute of Technology Gandhinagar, India                                                                                                                          |                                    |
| <b>Reviewer - ACM Transactions on Knowledge Discovery from Data</b>                                                                                                        | <b>June 2026</b>                   |
| <b>Reviewer - Empowering Global South AI @ AAAI 2026</b>                                                                                                                   | <b>December 2025</b>               |
| <b>Reviewer - GCLR @ AAAI 2026</b>                                                                                                                                         | <b>October 2025</b>                |
| <b>Reviewer - NeurIPS 2025</b>                                                                                                                                             | <b>July 2025</b>                   |
| <b>Ethics Reviewer - NeurIPS 2025</b>                                                                                                                                      | <b>July 2025</b>                   |
| <b>Ethics Reviewer - NeurIPS 2024</b>                                                                                                                                      | <b>August 2024</b>                 |
| <b>Ethics Reviewer - NeurIPS 2023</b>                                                                                                                                      | <b>July 2023</b>                   |
| <b>Volunteer - Foundations of Software Technology and Theoretical Computer Science</b>                                                                                     | <b>December 2024</b>               |
| Indian Institute of Technology Gandhinagar, India                                                                                                                          |                                    |
| <b>Volunteer - Randomized Methods For Approximation And Parameterized Algorithms</b>                                                                                       | <b>December 2022</b>               |
| Indian Institute of Technology Gandhinagar, India                                                                                                                          |                                    |
| <b>Volunteer - Indian Symposium on Machine Learning</b>                                                                                                                    | <b>December 2022</b>               |
| Indian Institute of Technology Gandhinagar, India                                                                                                                          |                                    |
| <b>Teaching Assistant - Scalable Data Science</b>                                                                                                                          | <b>July 2025 - November 2025</b>   |
| NPTEL                                                                                                                                                                      |                                    |
| <b>Teaching Assistant - Introduction to Data Science</b>                                                                                                                   | <b>January 2025 - May 2025</b>     |
| Indian Institute of Technology Gandhinagar, India                                                                                                                          |                                    |
| <b>Teaching Assistant - Introduction to Data Science</b>                                                                                                                   | <b>January 2023 - May 2023</b>     |
| Indian Institute of Technology Gandhinagar, India                                                                                                                          |                                    |
| <b>Teaching Assistant - Ordinary and partial differential equations</b>                                                                                                    | <b>August 2022 - November 2022</b> |
| Indian Institute of Technology Gandhinagar, India                                                                                                                          |                                    |

## Relevant Coursework

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|-------------------------------|------------------------|---------------------------------------|
| • Social and Complex Networks | • Optimisation in ML   | • Understanding Large Language Models |
| • Machine Learning            | • GNN and Transformers |                                       |

## Events Attended

|                                                                |                      |
|----------------------------------------------------------------|----------------------|
| <b>ACM SIGMOD/PODS</b>                                         | <b>May 2026</b>      |
| Bangalore, India                                               |                      |
| <b>Indian Symposium on Machine Learning</b>                    | <b>December 2025</b> |
| BITS Pilani Hyderabad Campus, India                            |                      |
| <b>Frontiers of Graph Algorithms</b>                           | <b>December 2025</b> |
| Indian Institute of Science, Bangalore, India                  |                      |
| <b>Neuroscience, Data Science and Dynamics</b>                 | <b>April 2025</b>    |
| International Center of Theoretical Sciences, Bangalore, India |                      |
| <b>Indian Symposium on Machine Learning</b>                    | <b>December 2024</b> |
| BITS Pilani Goa Campus, India                                  |                      |
| <b>Pingala Interactions in Computing 2024</b>                  | <b>February 2024</b> |
| Infosys Ltd Mysore, India                                      |                      |

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| <b>CODS-COMAD 2024</b><br>International Institute of Information Technology Bangalore, India                                                                                                                                                                                                                                                                                                                                  | <b>January 2024</b>  |
| <b>Indian Symposium on Machine Learning</b><br>Indian Institute of Technology Bombay, India                                                                                                                                                                                                                                                                                                                                   | <b>December 2023</b> |
| <b>Data Science: Probabilistic and Optimization Methods</b><br>International Center of Theoretical Sciences, Bangalore, India                                                                                                                                                                                                                                                                                                 | <b>July 2023</b>     |
| <b>Technical Skills</b>                                                                                                                                                                                                                                                                                                                                                                                                       |                      |
| <b>Languages:</b> Python, Java, C, C++, R, SQL, Shell, L <sup>A</sup> T <sub>E</sub> X<br><b>Technologies/Frameworks:</b> PyTorch, TensorFlow, GitHub<br><b>Communicating Language:</b> English, Hindi, Bengali                                                                                                                                                                                                               |                      |
| <b>Certificate and online Courses</b>                                                                                                                                                                                                                                                                                                                                                                                         |                      |
| <b>Certification in Scientific Writing</b><br>Indian Institute of Technology Gandhinagar                                                                                                                                                                                                                                                                                                                                      | <b>April 2025</b>    |
| <ul style="list-style-type: none"> <li>The Certificate in Scientific Writing is a professional credential that validates a participant's proficiency in writing clear, accurate, and well-structured scientific documents. It demonstrates the holder's ability to communicate research effectively in academic and professional settings.</li> </ul>                                                                         |                      |
| <b>Introduction to IoT</b><br>Cisco Networking Academy                                                                                                                                                                                                                                                                                                                                                                        | <b>August 2020</b>   |
| <ul style="list-style-type: none"> <li>Learned how the current digital transformation is creating unprecedented economic opportunity. Understood how the IoT is bridging the gap between operational and information technology systems. Discovered how standard business processes are being transformed.</li> </ul>                                                                                                         |                      |
| <b>Elements of AI</b><br>University of Helsinki                                                                                                                                                                                                                                                                                                                                                                               | <b>June 2020</b>     |
| <ul style="list-style-type: none"> <li>It provides a gentle introduction to the basics that enables us to understand what is possible, and what isn't, and to recognise how AI is affecting our lives.</li> <li>We can define and discuss AI, be critical about AI-related claims, explain the underlying technologies including machine learning and neural networks and understand the major implications of AI.</li> </ul> |                      |
| <b>Awards and Achievements</b>                                                                                                                                                                                                                                                                                                                                                                                                |                      |
| <b>Best Poster Award - PhD Research Showcase</b><br>Indian Institute of Technology Gandhinagar                                                                                                                                                                                                                                                                                                                                | <b>January 2025</b>  |
| <b>UGC-NET 2022</b><br>NTA UCG                                                                                                                                                                                                                                                                                                                                                                                                | <b>November 2022</b> |
| <ul style="list-style-type: none"> <li>Qualified for JRF &amp; Assistant Professor</li> </ul>                                                                                                                                                                                                                                                                                                                                 |                      |
| <b>GATE 2022</b>                                                                                                                                                                                                                                                                                                                                                                                                              | <b>March 2022</b>    |
| <b>Hacktoberfest 2022</b><br>Github and Digital Ocean                                                                                                                                                                                                                                                                                                                                                                         | <b>October 2022</b>  |
| <ul style="list-style-type: none"> <li>Successfully completed Hacktoberfest, 2022, by contributing in public Github repository.</li> </ul>                                                                                                                                                                                                                                                                                    |                      |
| <b>Hacktoberfest 2021</b><br>Github and Digital Ocean                                                                                                                                                                                                                                                                                                                                                                         | <b>October 2021</b>  |
| <ul style="list-style-type: none"> <li>Successfully completed Hacktoberfest, 2021, by contributing in public Github repository.</li> <li>Four pull requests were accepted by a public Github repository -  <b>Cosmos</b></li> </ul>                                                                                                        |                      |
| <b>Digital Wellness Online Challenge 2015</b><br>National e-governance Division                                                                                                                                                                                                                                                                                                                                               | <b>July 2015</b>     |